



ACADEMIC
PRESS

Available at
WWW.MATHEMATICSWEB.ORG
POWERED BY SCIENCE @ DIRECT®

Journal of Multivariate Analysis 85 (2003) 106–129

Journal of
Multivariate
Analysis

<http://www.elsevier.com/locate/jmva>

Multivariate nonparametric tests in a randomized complete block design

J. Möttönen,^{a,*} J. Hüsler,^b and H. Oja^c

^a *Department of Mathematical Sciences Statistics, University of Oulu, P.O. Box 3000, FIN-90014 University of Oulu, Finland*

^b *Universität Bern, Institut für Mathematische Statistik und Versicherungslehre, Sidlerstrasse 5, CH-3012 Bern, Switzerland*

^c *Department of Mathematics and Statistics, University of Jyväskylä, P.O. Box 35, FIN-40351 Jyväskylä, Finland*

Received 21 June 2001

Abstract

In this paper multivariate extensions of the Friedman and Page tests for the comparison of several treatments are introduced. Related unadjusted and adjusted treatment effect estimates for the multivariate response variable are also found and their properties discussed. The test statistics and estimates are analogous to the traditional univariate methods. In test constructions, the univariate ranks are replaced by multivariate spatial ranks (J. Nonparam. Statist. 5 (1995) 201). Asymptotic theory is developed to provide approximations for the limiting distributions of the test statistics and estimates. Limiting efficiencies of the tests and treatment effect estimates are found in the multivariate normal and t distribution cases. The tests are rotation invariant only, but affine invariant versions can be easily constructed. The theory is illustrated by an example.

© 2003 Elsevier Science (USA). All rights reserved.

AMS 1991 subject classifications: 62G10; 62G20; 62H15

Keywords: Affine invariance; Multivariate Friedman test; Multivariate Page test; Pitman efficiency; Rotation invariance; Spatial rank

*Corresponding author.

E-mail addresses: jyrki.mottonen@oulu.fi (J. Möttönen), juerg.huesler@stat.unibe.ch (J. Hüsler), ojahannu@cc.jyu.fi (H. Oja).

¹Supported by the Academy of Finland.

1. Introduction

The blocked design for the comparison of k treatments is obtained by generalizing the paired-sample design. This type of comparison is advisable if the variation among the subjects is high as compared to the variation caused by the treatments. In this approach, the subjects are randomly divided into homogeneous subgroups, called blocks, and the subject receiving different treatments are compared only within the blocks. In some cases, the treatments may even be compared on a single subject and the treatments are applied in a random order. A randomized complete block design with n blocks then requires blocks of size k ; the k subjects in each block are randomly assigned to the k treatments.

For univariate response variables, the most popular tests for considering the null hypothesis of no treatment effects are the regular balanced two-way analysis of variance (ANOVA) test and the nonparametric and (under the null hypothesis) distribution-free rank-based Friedman and Page tests. Naturally, the permutation test version of the ANOVA test is conditionally distribution-free as well. The Page test is a one-sided test in the sense that it is applicable when there is an a priori ordering of the treatment effects. For a discussion of the univariate tests and corresponding treatment difference estimates, see [4,5].

Our aim is to introduce new d -variate generalizations of the Friedman and Page tests with the companion estimates. There are several multivariate extensions of the sign and rank concepts based on different multivariate generalizations of the median. See for example [8] and references therein. Puri and Sen [11] extended the Friedman and Page tests just replacing the univariate ranks by the vectors of marginal ranks. In this paper, we use the so-called spatial (centered) ranks [6]: The spatial centered rank of the observation $\mathbf{y}_i$ among a d -variate data set $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$ is defined as

$$\mathbf{r}_i = \frac{1}{n} \sum_{j=1}^n \text{sgn}(\mathbf{y}_i - \mathbf{y}_j),$$

where

$$\text{sgn}(\mathbf{y}) = \begin{cases} \|\mathbf{y}\|^{-1} \mathbf{y} & \text{if } \mathbf{y} \neq \mathbf{0}, \\ \mathbf{0} & \text{if } \mathbf{y} = \mathbf{0} \end{cases}$$

is the spatial sign of $\mathbf{y}$ ($\|\mathbf{y}\| = (\mathbf{y}^T \mathbf{y})^{1/2}$ is the length of the vector $\mathbf{y}$). It is easy to see that the $\mathbf{r}_i$ are centered, that is, $\sum_{i=1}^n \mathbf{r}_i = \mathbf{0}$. Möttönen and Oja [6] used the spatial ranks to construct one-sample and several-samples multivariate analogs of the Wilcoxon test with corresponding Hodges–Lehmann-type location estimates.

The plan is as follows. In Section 2 the permutation MANOVA and multivariate Friedman and Page test statistics are introduced and their limiting permutational (conditional) distributions are found. Treatment difference estimates (adjusted and unadjusted) based on pairwise spatial sign tests are also introduced. In Section 3 the limiting unconditional distributions of the test statistic are derived under the null hypothesis as well as under sequences of contiguous alternatives. The Pitman efficiencies are then found for multivariate spherical normal and t distribution cases.

Also the efficiencies of the treatment effect estimates are discussed. In Section 4.1 it is shown how to construct affine invariant/equivariant versions of the tests and estimates. Section 4.2 serves as an example, a small robustness study and some final remarks. The proofs are postponed to the appendix.

2. Multivariate test statistics

2.1. Permutation MANOVA tests

We first state the problem. The data consist of $N = nk$ d -dimensional vectors. The $N = nk$ subjects are in n blocks and within each block the subjects are assigned to k treatments at random. The d -variate observations can then be presented in an $n \times k$ table as follows:

	Treatments			
Blocks	1	2	...	k
1	$\mathbf{y}_{11}$	$\mathbf{y}_{12}$	...	$\mathbf{y}_{1k}$
2	$\mathbf{y}_{21}$	$\mathbf{y}_{22}$	...	$\mathbf{y}_{2k}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
n	$\mathbf{y}_{n1}$	$\mathbf{y}_{n2}$	...	$\mathbf{y}_{nk}$

The $k \times d$ observed values in each block may also be concatenated as random vectors

$$\mathbf{y}_i = \begin{pmatrix} \mathbf{y}_{i1} \\ \vdots \\ \mathbf{y}_{ik} \end{pmatrix}, \quad i = 1, \dots, n.$$

It is then natural to assume that the $\mathbf{y}_i$ are independent with cdf's $F_i(\mathbf{y}_{i1} - \mathbf{A}_1, \dots, \mathbf{y}_{ik} - \mathbf{A}_k)$ where the F_i are *permutation invariant* and the $\mathbf{A}_j$, $j = 1, \dots, k$, $(\sum_j \mathbf{A}_j = \mathbf{0})$ are the treatment effects. By permutation invariance of F we mean that $F(\mathbf{y}_{i_1}, \dots, \mathbf{y}_{i_k}) = F(\mathbf{y}_1, \dots, \mathbf{y}_k)$ for all permutations $(i_1, \dots, i_k)$ of $(1, \dots, k)$. The treatment differences are given by

$$\mathbf{A}_{jj'} = \mathbf{A}_{j'} - \mathbf{A}_j, \quad j, j' = 1, \dots, k.$$

We want to test the hypothesis of no treatment effects, that is,

$$H_0: \mathbf{A}_1 = \dots = \mathbf{A}_k = \mathbf{0}.$$

As the cumulative distribution functions F_i , $i = 1, \dots, n$, are permutation invariant, the variables $\mathbf{y}_{i1}, \dots, \mathbf{y}_{ik}$ are, in each row i , exchangeable under the null model. Moreover, if $F_1 = \dots = F_n$, then the $\mathbf{y}_i$ are i.i.d.

Let

$$\mathbf{z}_i = \begin{pmatrix} \mathbf{z}_{i1} \\ \mathbf{z}_{i2} \\ \vdots \\ \mathbf{z}_{i,k-1} \end{pmatrix} = \begin{pmatrix} \mathbf{y}_{i1} - \bar{\mathbf{y}}_{i\cdot} \\ \mathbf{y}_{i2} - \bar{\mathbf{y}}_{i\cdot} \\ \vdots \\ \mathbf{y}_{i,k-1} - \bar{\mathbf{y}}_{i\cdot} \end{pmatrix}$$

denote the $(k-1)d$ -vector of the centered response values in the i th block. (In the centering, the blockwise mean vectors are used. The $\mathbf{z}_{i,k}$ is not needed in the concatenated vector as the sum of the centered values is $\mathbf{0}$.) As in the univariate case, the sum of the centered response values over the blocks

$$\mathbf{z} = \begin{pmatrix} \mathbf{z}_{\cdot 1} \\ \mathbf{z}_{\cdot 2} \\ \vdots \\ \mathbf{z}_{\cdot k-1} \end{pmatrix} = \sum_i \mathbf{z}_i$$

is used in test constructions.

Consider first, under the null hypothesis (no treatment effect), the conditional or *permutation distribution* of the table of observations and finally of vector $\mathbf{z}$. The corresponding permutation random variable is denoted by $\mathbf{z}^*$ with equiprobable values over all possible $(k!)^n$ values obtained by permuting within rows in the above table. Clearly,

$$P^*(\mathbf{z}_{i1}^* = \mathbf{z}_{ij_1}, \dots, \mathbf{z}_{ik}^* = \mathbf{z}_{ij_k} \mid \text{observed table}) = \frac{1}{k!}$$

for all permutations $(j_1, \dots, j_k)$ of $(1, \dots, k)$. Then the permutation expectations, variances and covariances of $\mathbf{z}_{ij}^*$ are

$$E^*(\mathbf{z}_{ij}^*) = \mathbf{0} \quad \text{and} \quad \text{Var}^*(\mathbf{z}_{ij}^*) = \frac{1}{k} \sum_{l=1}^k \mathbf{z}_{il} \mathbf{z}_{il}^T$$

and

$$\text{Cov}^*(\mathbf{z}_{ij}^*, \mathbf{z}_{i'j'}^*) = E^*(\mathbf{z}_{ij}^* \mathbf{z}_{i'j'}^{*T}) = -\frac{1}{k(k-1)} \sum_{l=1}^k \mathbf{z}_{il} \mathbf{z}_{il}^T \quad \text{for } j \neq j'.$$

Naturally, $\mathbf{z}_{ij}^*$ and $\mathbf{z}_{i'j'}^*$ are independent if $i \neq i'$. Finally,

$$E^*(\mathbf{z}^*) = \mathbf{0} \quad \text{and} \quad S = \text{Var}^*(\mathbf{z}^*) = \frac{n}{k-1} (kI_{k-1} - J_{k-1}) \otimes C,$$

where J_k is a $k \times k$ matrix of ones and

$$C = \frac{1}{nk} \sum_{i=1}^n \sum_{j=1}^k \mathbf{z}_{ij} \mathbf{z}_{ij}^T.$$

Note that, if C is of full rank, also S is of full rank, $d(k-1)$, and its inverse is

$$S^{-1} = \frac{k-1}{nk} (I_{k-1} + J_{k-1}) \otimes C^{-1}$$

(see e.g. [13]).

Now we can define a squared form test statistic for testing H_0 and give its limiting permutational distribution.

Definition 2.1. MANOVA test statistic for testing H_0 is

$$Q = \mathbf{z}^T S^{-1} \mathbf{z} = \frac{k-1}{nk} \sum_{j=1}^k \mathbf{z}_{\cdot j}^T C^{-1} \mathbf{z}_{\cdot j}.$$

Theorem 2.1. Assume that the sequence $(\mathbf{y}_i)$ is such that the second and third moments of $\mathbf{z}_i^*$ are uniformly bounded with $C \rightarrow \Sigma$ where Σ is of full rank. Then, under the null hypothesis, the limiting permutation distribution of $Q^* = \mathbf{z}^* T S^{-1} \mathbf{z}^*$ is $\chi_{d(k-1)}^2$.

Note that if we write

$$F = n \sum_{j=1}^k (\bar{\mathbf{y}}_{\cdot j} - \bar{\mathbf{y}}_{\cdot \cdot})(\bar{\mathbf{y}}_{\cdot j} - \bar{\mathbf{y}}_{\cdot \cdot})^T,$$

then the statistic

$$Q = \frac{k-1}{k} \text{tr}(FC^{-1})$$

is the Pillai's trace statistic. The statistic Q is naturally affine invariant.

2.2. Spatial rank tests

Permutation rank tests can be derived in exactly the same way as the permutation MANOVA tests in Section 2.1. The multivariate blockwise centered response vectors $\mathbf{z}_{ij}$ are just replaced by multivariate blockwise centered rank vectors. We thus compute, within the blocks, the multivariate centered ranks: r_{ij} is the centered rank of the observation with the j th treatment among the observations in the i th block. The ranks can be displayed in a table as follows:

	Treatments				
Blocks	1	2	...	k	Σ
1	r_{11}	r_{12}	...	r_{1k}	$\mathbf{0}$
2	r_{21}	r_{22}	...	r_{2k}	$\mathbf{0}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
n	r_{n1}	r_{n2}	...	r_{nk}	$\mathbf{0}$
Σ	$r_{\cdot 1}$	$r_{\cdot 2}$	...	$r_{\cdot k}$	$\mathbf{0}$

Let $\mathbf{r}_i = (\mathbf{r}_{i1}^T, \dots, \mathbf{r}_{i,k-1}^T)^T$ denote the concatenated vector of rank vectors in the i th block ($\mathbf{r}_{ik} = -\sum_{j=1}^{k-1} \mathbf{r}_{ij}$) and let $\mathbf{r} = (\mathbf{r}_1^T, \dots, \mathbf{r}_{k-1}^T)^T = \sum_i \mathbf{r}_i$ be the vector of the rank sums over the blocks.

The permutation distribution of vector $\mathbf{r}$ under the null hypothesis can be easily derived since the observations $\mathbf{y}_{i1}, \dots, \mathbf{y}_{ik}$ as well as the rank vectors $\mathbf{r}_{i1}, \dots, \mathbf{r}_{ik}$ are exchangeable within the block (due to the randomization). Let $\mathbf{r}_{ij}^*$ and $\mathbf{r}^*$ denote the permutation random variables of $\mathbf{r}_{ij}$ and $\mathbf{r}$, correspondingly. The permutation expectations and covariance matrices are now

$$E^*(\mathbf{r}_{ij}^*) = \mathbf{0} \quad \text{and} \quad \text{Var}^*(\mathbf{r}_{ij}^*) = \frac{1}{k} \sum_{l=1}^k \mathbf{r}_{il} \mathbf{r}_{il}^T$$

and

$$\text{Cov}^*(\mathbf{r}_{ij}^*, \mathbf{r}_{ij'}^*) = -\frac{1}{k(k-1)} \sum_{l=1}^k \mathbf{r}_{il} \mathbf{r}_{il}^T \quad \text{for } j \neq j',$$

and finally

$$E^*(\mathbf{r}^*) = \mathbf{0} \quad \text{and} \quad S_r = \text{Var}^*(\mathbf{r}^*) = \frac{n}{k-1} (kI_{k-1} - J_{k-1}) \otimes C_r$$

with

$$C_r = \frac{1}{nk} \sum_{i=1}^n \sum_{j=1}^k \mathbf{r}_{ij} \mathbf{r}_{ij}^T.$$

The inverse of the permutation covariance matrix is

$$S_r^{-1} = \frac{k-1}{nk} (I_{k-1} + J_{k-1}) \otimes C_r^{-1}.$$

Definition 2.2. (A) Multivariate Friedman test statistic is

$$Q_r = \mathbf{r}^T S_r^{-1} \mathbf{r} = \frac{k-1}{nk} \sum_{j=1}^k \mathbf{r}_j^T C_r^{-1} \mathbf{r}_j.$$

(B) Multivariate Page test statistic is

$$\mathbf{L} = \mathbf{r}_{\cdot 1} + 2\mathbf{r}_{\cdot 2} + \dots + k\mathbf{r}_{\cdot k} = \sum_{j=1}^k j\mathbf{r}_j.$$

Note that Q_r and $\mathbf{L}$ are rotations but not scale-invariant/-equivariant. As in the univariate case, the Page test is constructed to detect alternatives with an ordering in the effects of the treatments: There is a ‘direction’ $\mathbf{u}$ (corresponding to a linear combination of the original response variables) such that

$$\mathbf{u}^T \mathbf{A}_1 \leq \mathbf{u}^T \mathbf{A}_2 \leq \dots \leq \mathbf{u}^T \mathbf{A}_k.$$

Theorem 2.2. Assume that the null hypothesis of no treatment effect is true and that the sequence $(\mathbf{y}_i)$ is chosen so that $C_r \rightarrow \Sigma_r$ where Σ_r is of full rank.

(A) The limiting permutation distribution of $Q_r^* = \mathbf{r}^* T S_r^{-1} \mathbf{r}^*$ is chi square distribution with $d(k-1)$ degrees of freedom.

(B) The limiting permutation distribution of

$$P^* = \frac{12}{nk^2(k+1)} \mathbf{L}^* T C_r^{-1} \mathbf{L}^*$$

is a chi square distribution with d degrees of freedom.

Note that

$$E^*(\mathbf{L}^*) = \mathbf{0} \quad \text{and} \quad \text{Var}^*(\mathbf{L}^*) = \frac{nk^2(k+1)}{12} C_r.$$

It is easy to see that the test statistics Q_r and $\mathbf{L}$ reduce to classical Friedman and Page test statistics as $d = 1$.

2.3. Pairwise tests and treatment effect estimates

Consider now pairwise spatial sign test statistics,

$$\mathbf{T}_{jj'} = \sum_{i=1}^n \text{sgn}(\mathbf{y}_{ij'} - \mathbf{y}_{ij}), \quad j, j' = 1, \dots, k$$

for comparing treatments j and j' [6] and let

$$U = (\mathbf{T}_{jj'}), \quad j, j' = 1, \dots, k$$

be the $dk \times k$ matrix of observed test statistic values. Note that $\mathbf{T}_{jj} = \mathbf{0}$ and $\mathbf{T}_{jj'} = -\mathbf{T}_{j'j}$, $j, j' = 1, \dots, k$.

The following lemma implies that the multivariate Friedman and Page tests may be derived from these pairwise sign tests.

Lemma 2.1. $(\mathbf{r}_1^T, \dots, \mathbf{r}_k^T)^T = -\frac{1}{k} U \mathbf{1}_k$.

Next, note that the companion treatment difference estimates $\hat{\Delta}_{jj'}$ to pairwise sign tests $\mathbf{T}_{jj'}$ are the spatial medians of the differences $\mathbf{y}_{ij'} - \mathbf{y}_{ij}$, $i = 1, \dots, n$. Let again

$$\hat{\Delta} = (\hat{\Delta}_{jj'}), \quad j, j' = 1, \dots, k$$

be the $dk \times k$ matrix of treatment difference estimates ($\hat{\Delta}_{jj} = 0$, $j = 1, \dots, k$). As in the univariate case, the problem then is that these estimates may not be consistent in the sense that $\hat{\Delta}_{jj''} = \hat{\Delta}_{jj'} + \hat{\Delta}_{j'j''}$. A solution to this consistency problem is obtained as follows. First, take the averages

$$\hat{\Delta}_j = -\frac{1}{k} \sum_{j'=1}^k \hat{\Delta}_{jj'}.$$

Then the adjusted treatment difference estimates are $\tilde{\Delta}_{jj'} = \hat{\Delta}_{j'} - \hat{\Delta}_j$ and the matrix of these adjusted estimates is then

$$\tilde{\Delta} = (\tilde{\Delta}_{jj'}), \quad j, j' = 1, \dots, k.$$

For univariate case, see [4,5]. For multivariate treatment estimates based on marginal sign and rank tests, see [11].

3. Limiting distributions and efficiency

For considering asymptotic properties of the tests and estimates we relax the assumptions and assume that the original observations y_{ij} are independent and that the cdf of the y_{ij} is $F(y - \theta_i - \Delta_j)$ where the θ_i are the block effects and the Δ_j the treatment effects ($\sum_i \theta_i = \sum_j \Delta_j = \mathbf{0}$). We wish to test the hypothesis

$$H_0 : \Delta_1 = \dots = \Delta_k = \mathbf{0} \text{ versus } H_n : \Delta_j = \frac{1}{\sqrt{n}} \delta_j, \quad j = 1, \dots, k$$

where also $\sum_j \delta_j = \mathbf{0}$. Let δ denote the vector $(\delta_1^T \dots \delta_{k-1}^T)^T$.

Under the null hypothesis, $y_{i1}, \dots, y_{ik}$ are exchangeable. This further implies that, again under the null hypothesis, the rank vectors $r_{i1}, \dots, r_{ik}$ are exchangeable. Moreover, the $y_i - \theta_i$ as well as the r_i are i.i.d.

3.1. Tests for comparing two treatments ($k = 2$)

For limiting distributions, we first recall the asymptotic theory for spatial sign and rank tests for comparing two treatments. This is done also to introduce matrices A_1 , B_1 , A_2 and B_2 needed in the subsequent discussion. First, we consider the dependent samples case (matched pairs) and use the one-sample spatial sign test. The one-sample spatial sign test statistic [6] for the difference vectors $y_{i2} - y_{i1}$, $i = 1, \dots, n$, for example, is defined as

$$T_{12} = \sum_{i=1}^n \text{sgn}(y_{i2} - y_{i1}).$$

Using the results of Möttönen and Oja [6] we get, under H_n ,

$$n^{-1/2} T_{12} \xrightarrow{d} N_d(A_1(\delta_2 - \delta_1), B_1),$$

where

$$B_1 = E_0(\text{sgn}(y_{i2} - y_{i1}) \text{sgn}(y_{i2} - y_{i1})^T)$$

is the spatial sign covariance matrix for difference vectors and, independently of n ,

$$A_1 = \nabla E_{A_{12}} \left[\frac{1}{n} T_{12} \right] \Big|_{A_{12}=\mathbf{0}} = \nabla E_{A_{12}} [\text{sgn}(y_{i2} - y_{i1})] \Big|_{A_{12}=\mathbf{0}},$$

where ∇ denotes differentiation with respect to the components of the treatment difference $\mathbf{A}_{12}$. See Section 6.1 in [2].

For a moment assume that there is no block effect and consequently $\mathbf{y}_{11}, \dots, \mathbf{y}_{n1}$ and $\mathbf{y}_{12}, \dots, \mathbf{y}_{n2}$ are two independent random samples. One should then use the two-sample spatial rank test. Let

$$\mathbf{q}_N(\mathbf{y}_{ij}) = \frac{1}{N} \sum_{i'=1}^n \sum_{j'=1}^2 \text{sgn}(\mathbf{y}_{ij} - \mathbf{y}_{i'j'})$$

be the spatial rank vector of $\mathbf{y}_{ij}$ among $N = 2n$ observations coming from treatments 1 and 2. Then the two-sample spatial rank test statistic [6] for testing the null hypothesis $H_0: \mathbf{A}_{12} = \mathbf{0}$ is

$$\mathbf{S}_{12} = \sum_{i=1}^n \mathbf{q}_N(\mathbf{y}_{i2}).$$

Möttönen and Oja [6] gives us the result, under H_n ,

$$N^{-1/2} \frac{1}{4} \mathbf{S}_{12} \xrightarrow{d} N_d(A_2(\boldsymbol{\delta}_2 - \boldsymbol{\delta}_1), \frac{1}{4} B_2),$$

where

$$B_2 = E_0(\text{sgn}(\mathbf{y}_{ij} - \mathbf{y}_{i'j'}) \text{sgn}(\mathbf{y}_{ij} - \mathbf{y}_{i''j''})^T)$$

is the spatial rank covariance matrix [15] and

$$A_2 = \nabla E_{A_{12}} \frac{1}{N} \frac{1}{4} \mathbf{S}_{12} \Big|_{A_{12}=\mathbf{0}} = \nabla E_{A_{12}} [\text{sgn}(\mathbf{y}_{i2} - \mathbf{y}_{i1})] \Big|_{A_{12}=\mathbf{0}}$$

which equals A_1 . From this on we denote $A = A_1 = A_2$.

3.2. MANOVA, Friedman and Page tests

We start by giving, as a corollary of the theorem below, the unconditional limiting distribution of the MANOVA test statistic under contiguous sequences of alternatives.

Theorem 3.1. *Under the alternative sequence H_n , the limiting distribution of $n^{-1/2} \mathbf{z}$ is a $d(k-1)$ -variate normal distribution with mean vector $\boldsymbol{\delta}$ and covariance matrix $\frac{1}{k-1} (kI_{k-1} - J_{k-1}) \otimes \Sigma$ where $\Sigma = E_0(\mathbf{z}_{ij} \mathbf{z}_{ij}^T) = \frac{k-1}{k} E_0(\mathbf{y}_{ij} \mathbf{y}_{ij}^T)$ is the covariance matrix of $\mathbf{z}_{ij}$.*

Corollary 3.1. *Under the sequence of alternatives H_n , the limiting distribution of the MANOVA test statistic Q is a noncentral χ^2 distribution with $d(k-1)$ degrees of freedom and a noncentrality parameter*

$$\delta_{\text{MANOVA}}^2 = \frac{k-1}{k} \sum_{j=1}^k \boldsymbol{\delta}_j^T \Sigma^{-1} \boldsymbol{\delta}_j,$$

Now we are ready to present limiting distributions for multivariate Friedman and Page tests. The gradients of the mean vectors $A = A_1 = A_2$ as well as spatial sign and rank covariance matrices B_1 and B_2 introduced in Section 3.1 are utilized here. We also write

$$B = E_0(\mathbf{r}_{ij}\mathbf{r}_{ij}^T) = \frac{k-1}{k^2}B_1 + \frac{(k-1)(k-2)}{k^2}B_2$$

for the covariance matrix of $\mathbf{r}_{ij}$.

Theorem 3.2. *Under the alternative sequence H_n , the limiting distribution of $n^{-1/2}\mathbf{r}$ is a $d(k-1)$ -variate normal distribution with mean vector $(I_{k-1} \otimes A)\boldsymbol{\delta}$ and covariance matrix $\frac{1}{k-1}(kI_{k-1} - J_{k-1}) \otimes B$.*

In practice, the matrix B is, of course, unknown but can be (under the null hypothesis) consistently estimated by C_r .

Corollary 3.2. *Under the sequence of alternatives H_n ,*

(A) *the limiting distribution of the Friedman test statistic Q_r is a noncentral χ^2 distribution with $d(k-1)$ degrees of freedom and a noncentrality parameter*

$$\delta_{\text{FRIEDMAN}}^2 = \frac{k-1}{k} \sum_{j=1}^k \boldsymbol{\delta}_j^T A^T B^{-1} A \boldsymbol{\delta}_j;$$

and

(B) *the limiting distribution of the Page test statistic P is a noncentral χ^2 distribution with d degrees of freedom and noncentrality parameter*

$$\delta_{\text{PAGE}}^2 = \frac{12}{k^2(k+1)} \left(\sum_j j \boldsymbol{\delta}_j^T \right) A^T B^{-1} A \left(\sum_j j \boldsymbol{\delta}_j \right).$$

Relative efficiencies for comparing the first two tests are simply given by

Theorem 3.3. *The Pitman asymptotic relative efficiency of the multivariate Friedman test with respect to MANOVA is*

$$\text{ARE}_{12} = \frac{\delta_{\text{FRIEDMAN}}^2}{\delta_{\text{MANOVA}}^2} = \frac{\sum_j \boldsymbol{\delta}_j^T A^T B^{-1} A \boldsymbol{\delta}_j}{\sum_j \boldsymbol{\delta}_j^T \Sigma^{-1} \boldsymbol{\delta}_j}.$$

Note that $B = \frac{k-1}{k^2}B_1 + \frac{(k-1)(k-2)}{k^2}B_2$ and the Pitman efficiency of Friedman test is that of the pairwise spatial sign test as $k=2$ ($B = \frac{1}{4}B_1$) and goes towards the efficiency of the spatial rank test as $k \rightarrow \infty$ ($B \rightarrow B_2$). This extends the well-known univariate result (7.2.42) in [11].

If the observations come from $N_d(\mathbf{0}, I_d)$,

$$\Sigma = \frac{k-1}{k}I_d, \quad A = \frac{\Gamma(\frac{d+1}{2})}{2\Gamma(\frac{d+2}{2})}I_d, \quad B_1 = \frac{1}{d}I_d$$

and

$$B_2 = \frac{1}{4} \frac{\Gamma^2(\frac{d+1}{2})}{\Gamma^2(\frac{d+2}{2})} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; \frac{d+2}{2}; \frac{1}{4}\right) I_d,$$

where ${}_2F_1$ is the Gauss hypergeometric function. See Möttönen et al. (1995). The corollary then follows.

Corollary 3.3. *For the $N_d(\mathbf{0}, I_d)$ distribution,*

$$\text{ARE}_{12} = \left[\frac{d\Gamma^2(\frac{d}{2})}{k\Gamma^2(\frac{d+1}{2})} + \left(1 - \frac{2}{k}\right) {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; \frac{d+2}{2}; \frac{1}{4}\right) \right]^{-1}$$

and $\lim_{d \rightarrow \infty} \text{ARE}_{12} = 1$.

If the observations come from the multivariate t distribution with v degrees of freedom,

$$\Sigma = \frac{(k-1)v}{k(v-2)} I_d, \quad A = \frac{\Gamma(\frac{2v+1}{2})\Gamma^2(\frac{v+1}{2})\Gamma(\frac{d+1}{2})}{\sqrt{v}\Gamma^2(\frac{v}{2})\Gamma(\frac{d+2}{2})v!} I_d, \quad B_1 = \frac{1}{d} I_d$$

and

$$B_2 = \frac{\Gamma(\frac{d+v}{2})\Gamma(\frac{3v}{2})\Gamma^2(\frac{d+1}{2})\Gamma^2(\frac{v+1}{2})}{2\Gamma^3(\frac{v}{2})\Gamma^2(\frac{d+2}{2})\Gamma(\frac{d+3v+2}{2})} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{(\frac{d+1}{2})_i (\frac{v+1}{2})_i (\frac{d+1}{2})_j (\frac{v+1}{2})_j (\frac{d+2}{2})_{i+j}}{(\frac{d+2}{2})_i (\frac{d+2}{2})_j (\frac{d+3v+2}{2})_{i+j} i! j!} I_d \\ \doteq b_2 I_d,$$

where $(a)_i = \Gamma(a+i)/\Gamma(a)$ Möttönen et al. [7]. Note that b_2 is a constant times a hypergeometric series of two variables (Kampé de Fériet function).

Corollary 3.4. *For $t_{v,d}$ distribution,*

$$\text{ARE}_{12} = \frac{\Gamma^2(\frac{2v+1}{2})\Gamma^4(\frac{v+1}{2})\Gamma^2(\frac{d+1}{2})}{(v-2)\Gamma^4(\frac{v}{2})\Gamma^2(\frac{d+2}{2})(v!)^2} \left[\frac{1}{kd} + \left(1 - \frac{2}{k}\right) b_2 \right]^{-1}$$

and

$$\lim_{d \rightarrow \infty} \text{ARE}_{12} = \frac{2k\Gamma^2(\frac{2v+1}{2})\Gamma^4(\frac{v+1}{2})}{(v-2)\Gamma^4(\frac{v}{2})(v!)^2} \\ \times \left[1 + \frac{(k-2)\Gamma(\frac{3v}{2})}{2^{v/2}\Gamma^3(\frac{v}{2})} \int_0^\infty \int_0^\infty \frac{(uv)^{\frac{v-1}{2}}}{(\frac{1}{2}+u)^{\frac{1}{2}}(\frac{1}{2}+v)^{\frac{1}{2}}(\frac{1}{2}+u+v)^{\frac{3v}{2}}} du dv \right]^{-1}.$$

Next, we find the formulae for the Pitman asymptotic relative efficiency of the multivariate Page test with respect to Friedman test. As the degrees of freedom in the limiting distributions do not match any more, a direct use of the ratio of the noncentrality parameters is not valid. Nyblom and Mäkeläinen [9] showed how to extend the definition of the Pitman asymptotic relative efficiency to the cases where

the limiting distributions of the competing test statistics are of different types. The obtained efficiency may then depend on the alternative sequence, the chosen limiting size, and the power. In our case, the next Theorem then directly follows.

Theorem 3.4. *The Pitman asymptotic relative efficiency of the multivariate Page test with respect to the multivariate Friedman test is*

$$\text{ARE}_{23} = \frac{\gamma(d(k-1), \alpha, \beta)}{\gamma(d, \alpha, \beta)} \frac{\delta_{\text{PAGE}}^2}{\delta_{\text{FRIEDMAN}}^2},$$

where α and β are the chosen limiting power and size of the tests and $\gamma(d, \alpha, \beta)$ and $\gamma(d(k-1), \alpha, \beta)$ are given by

$$P(\chi_d^2(0) > \chi_{d,1-\alpha}^2) = P(\chi_{d(k-1)}^2(0) > \chi_{d(k-1),1-\alpha}^2) = \alpha$$

and

$$P(\chi_d^2(\gamma(d, \alpha, \beta)) > \chi_{d,1-\alpha}^2) = P(\chi_{d(k-1)}^2(\gamma(d(k-1), \alpha, \beta)) > \chi_{d(k-1),1-\alpha}^2) = \beta.$$

The efficiencies in the spherical multivariate t distribution case with selected dimensions d and degrees of freedom ν ($\nu = \infty$ gives the spherical multivariate normal distribution) are provided in Tables 1 and 2. Table 1 lists efficiencies of the Friedman test with respect to the regular MANOVA test. The figures show that the Friedman test clearly outperforms the classical MANOVA test for heavy tailed distributions. The efficiencies increase with the dimension d as well as with the number of treatments k . In the multinormal case the efficiency goes to one as dimension $d \rightarrow \infty$. Table 2 shows the high efficiency of the Page test against the alternatives where the effects of treatments are naturally ordered.

3.3. Treatment effect estimates

For treatment effect estimates, consider again first pairwise spatial sign test statistics $\mathbf{T}_{jj'}$ introduced in Section 2.3. In Section 3.1 we recalled that, under H_n ,

$$n^{-1/2} \mathbf{T}_{jj'} \xrightarrow{d} N_d(A_1(\boldsymbol{\delta}_{j'} - \boldsymbol{\delta}_j), B_1).$$

Recall also that we denote $A = A_1 = A_2$. See Section 2 in Möttönen et al. (1995) and Section 2.1 in [6]. Write

$$U = (\mathbf{T}_{jj'}) \text{ and } M = (A(\boldsymbol{\delta}_{j'} - \boldsymbol{\delta}_j)), \quad j, j' = 1, \dots, k.$$

In the following, the variance–covariance structure of a $dk \times k$ matrix valued random variable U is given as

$$\text{Var}(U) = E((U - E(U)) \otimes (U - E(U))^T).$$

The matrix $\text{Var}(U)$ is thus a $k^2 d \times k^2 d$ matrix with a block hierarchy of two levels: The $j'j'$ th element of the ij th block is the covariance matrix of $\mathbf{T}_{ij}$ and $\mathbf{T}_{ij'}$, say

Table 1
Asymptotic relative efficiencies of the spatial multivariate Friedman tests with respect to the classical MANOVA in the spherical multivariate $t(v)$ distribution case

Dimension (d)	DF (v)	Number of treatments (k)				
		2	3	5	10	∞
1	4	0.934	1.051	1.167	1.274	1.401
	10	0.702	0.790	0.878	0.958	1.054
	∞	0.637	0.716	0.796	0.868	0.955
2	4	1.152	1.233	1.306	1.367	1.434
	10	0.867	0.925	0.979	1.023	1.071
	∞	0.785	0.838	0.885	0.924	0.967
3	4	1.245	1.308	1.362	1.406	1.453
	10	0.937	0.980	1.018	1.048	1.081
	∞	0.849	0.887	0.919	0.946	0.973
5	4	1.328	1.373	1.412	1.442	1.473
	10	0.999	1.028	1.053	1.072	1.092
	∞	0.905	0.929	0.949	0.965	0.981
10	4	1.396	1.427	1.452	1.472	1.493
	10	1.050	1.067	1.081	1.092	1.103
	∞	0.951	0.964	0.974	0.981	0.989
∞	4	1.467	1.483	1.497	1.507	1.517
	10	1.103	1.108	1.112	1.115	1.118
	∞	1.000	1.000	1.000	1.000	1.000

Table 2
Asymptotic relative efficiency of multivariate Page test with respect to multivariate Friedman test in the spherical multivariate normal distribution case. $\alpha = 0.05$, $\beta = 0.5$ and $\delta = ((1 - k)\mathbf{1}_d^T, (3 - k)\mathbf{1}_d^T, \dots, (k - 3)\mathbf{1}_d^T)^T$

Dimension (d)	Number of treatments (k)				
	2	3	5	10	∞
1	1.000	1.290	1.671	2.294	∞
2	1.000	1.295	1.696	2.366	∞
3	1.000	1.303	1.719	2.422	∞
5	1.000	1.315	1.754	2.499	∞
10	1.000	1.334	1.803	2.603	∞
∞	1.000	1.414	2.000	3.000	∞

$\Sigma_{(ij),(i'j')}$. For a similar representation, see [12]. We now collect the results in the following two theorems. The proof of the first theorem easily follows from the results listed in Section 3.1.

Theorem 3.5. *Under the alternative sequence H_n , the limiting distribution of $n^{-1/2}U$ is a multivariate singular normal distribution with mean matrix M and covariance matrix determined by*

$$\Sigma_{(ij),(ij)} = B_1, \quad \Sigma_{(ij),(il)} = B_2, \quad \text{and} \quad \Sigma_{(ij),(lm)} = 0,$$

for $i < j < l < m$.

The first part of the proof of the next theorem is as of Theorem 6.1.2. in [2]. The second part is also straightforward as the $\tilde{A}_{ij}$ are linear combinations of the $\hat{A}_{ij}$.

Let then $\hat{A}$ and $\tilde{A}$ be the treatment difference estimates ($kd \times k$ matrices) introduced in Section 2.3. For the assumptions of the theorem see [2, Definition 6.1.3].

Theorem 3.6. *Under general assumptions, the limiting distribution of $\sqrt{n}(\hat{A} - A)$ is a multivariate singular normal with mean matrix zero and covariance matrix given by*

$$\Sigma_{(ij),(ij)} = A^{-1}B_1A^{-1}, \quad \Sigma_{(ij),(il)} = A^{-1}B_2A^{-1}, \quad \text{and} \quad \Sigma_{(ij),(lm)} = 0,$$

for $i < j < l < m$. Moreover, the limiting distribution of $\sqrt{n}(\tilde{A} - A)$ is a multivariate singular normal with mean matrix zero and covariance matrix given by

$$\Sigma_{(ij),(ij)} = \frac{2k}{k-1}A^{-1}BA^{-1}, \quad \Sigma_{(ij),(il)} = \frac{k}{k-1}A^{-1}BA^{-1}, \quad \text{and} \quad \Sigma_{(ij),(lm)} = 0,$$

for $i < j < l < m$.

The asymptotic relative efficiencies between competing treatment difference estimates is usually given in terms of the Wilks generalized variance: The asymptotic relative efficiency (ARE) of $\tilde{A}_{jj'}$ with respect to $\hat{A}_{jj'}$ is then given by (see Table 3)

$$\left(\frac{|A^{-1}B_1A^{-1}|}{|\frac{2k}{k-1}A^{-1}BA^{-1}|} \right)^{1/d}.$$

On the other hand, the asymptotic relative efficiency of $\tilde{A}_{jj'}$ with respect to $\bar{y}_{\cdot j'} - \bar{y}_{\cdot j}$ is

$$\left(\frac{|\Sigma|}{|A^{-1}BA^{-1}|} \right)^{1/d}.$$

This expands the univariate result (7.4.25) in [11]. Note that when the observations come from a spherical distribution the asymptotic relative efficiency of the estimate $\tilde{A}_{jj'}$ with respect to $\hat{A}_{jj'}$ is the same as the asymptotic relative efficiency of the multivariate Friedman test in Theorem 3.3.

Table 3

Asymptotic relative efficiency of $\hat{\Delta}_{jj'}$ with respect to $\hat{\Delta}_{jj'}$ in the spherical multinormal distribution case

Dimension (d)	Number of treatments (k)				
	2	3	5	10	∞
1	1.000	1.125	1.250	1.364	1.500
2	1.000	1.067	1.127	1.176	1.231
3	1.000	1.045	1.083	1.114	1.147
5	1.000	1.026	1.049	1.066	1.084
10	1.000	1.013	1.024	1.032	1.040
∞	1.000	1.000	1.000	1.000	1.000

4. Practical notions

4.1. Affine invariant versions of the tests

The tests discussed above are rotation invariant but not scale invariant. In some applications (e.g. directional data), only rotational invariance is needed. Due to the lack of scale invariance property, rescaling of one of the response variables may however change the results (e.g. p -values) and also highly reduce the efficiency of the tests and estimates. Next, we show how the transformation and retransformation approach introduced by Chakraborty and Chaudhury [1] can be easily used to construct affine invariant tests and affine equivariant estimates. The efficiencies of the procedures are then as in the spherical case.

Let D be any scatter matrix calculated from the data $\mathbf{y}_{ij}$. Then D is a positive definite symmetric $d \times d$ matrix and it is affine equivariant in the sense that if $D^\dagger$ is calculated from the transformed data $\mathbf{y}_{ij}^\dagger = G\mathbf{y}_{ij}$, $i = 1, \dots, n$ and $j = 1, \dots, k$, then $D^\dagger = GDG^T$. In our case, the transformation and retransformation approach then works as follows: (1) Construct the tests based on the transformed data $D^{-1/2}\mathbf{y}_{ij}$. This test is affine invariant. (2) Find the treatment difference estimates $\hat{\Delta}_{jj'}$ for the transformed data $D^{-1/2}\mathbf{y}_{ij}$. The final estimates are the ‘retransformed’ $D^{1/2}\hat{\Delta}_{jj'}$.

The invariance and equivariance properties of the above procedures are easily seen. As earlier, let $\mathbf{y}_{ij}^\dagger = G\mathbf{y}_{ij}$, and $D^\dagger = GDG^T$. First note that

$$D^{\dagger^{-1/2}}\mathbf{y}_{ij}^\dagger = D^{\dagger^{-1/2}}GD^{1/2}D^{-1/2}\mathbf{y}_{ij} = P[D^{-1/2}\mathbf{y}_{ij}],$$

where $P = D^{\dagger^{-1/2}}GD^{1/2}$ is an orthogonal $d \times d$ matrix. As the tests are rotation invariant, the results from the data set $D^{\dagger^{-1/2}}\mathbf{y}_{ij}^\dagger$ and from the data set $D^{-1/2}\mathbf{y}_{ij}$ are identical. Second, let the $\hat{\Delta}_{jj'}$ be the treatment difference estimates for the transformed data set $D^{-1/2}\mathbf{y}_{ij}$. By rotation equivariance property of the spatial median, the treatment difference estimates for the data set $D^{\dagger^{-1/2}}\mathbf{y}_{ij}^\dagger$ are $P\hat{\Delta}_{jj'}$. Retransforming $P\hat{\Delta}_{jj'}$ yields

$$D^{+1/2} P \hat{A}_{jj'} = G[D^{1/2} \hat{A}_{jj'}]$$

and also the affine equivariance follows.

Under general assumptions, if D is a $\sqrt{n}$ -convergent estimate of Σ , the limiting distributions of the test statistics based on the data set $D^{-1/2} \mathbf{y}_{ij}$ and the data set $\Sigma^{-1/2} \mathbf{y}_{ij}$ are the same (under the null hypothesis as well as under contiguous sequences of alternatives). This together with the above notions imply that the asymptotic relative efficiencies of the affine invariant versions of the tests in the elliptic multinormal and multivariate t distribution cases coincide with those in the respective spherical case (listed above). Finally, note that also shape matrices (e.g. Tyler's [14] estimate; the shape matrix differs from the scatter matrix in that the trace of the shape matrix is set to be d) may be used for the transformation and retransformation steps. This is due to the fact that the test is invariant under rescaling the variables with the same constant. For simultaneous location and shape estimation, see also [3].

4.2. An example and final remarks

We now illustrate the use of multivariate Friedman test on a data set earlier analyzed by Seber [13]. A randomized complete block design experiment was arranged to study the effects of six different treatments on plots of bean plants infested by the serpentine leaf miner insect. In this study the number of treatments was $k = 6$ and the number of blocks was $n = 4$. The measurement vectors consist of three different variables (from top to bottom): (1) the number of miners per leaf, (2) the weight of beans per plot (in kilograms) and (3) $\sin^{-1}(\sqrt{p})$, where p is the proportion of leaves infested with borer. See Table 4 for the original data set and Table 5 for blockwise centered rank vectors. The data set were analyzed using the permutation MANOVA test, multivariate Friedman test and an affine invariant version of the multivariate Friedman test. As mentioned in the previous Section 4.1, we can construct affine invariant tests by using shape matrices. In our case a natural way to do this is to find a transformation matrix W for which the rank covariance matrix C_r of the transformed data vectors $W \mathbf{y}_{ij}$ is a constant times identity matrix. Invariant tests and equivariant estimates are then obtained using the following iterative procedure:

1. Set $W \leftarrow I_d$
2. Calculate C_r using the transformed observations $W \mathbf{y}_{ij}$.
3. Set $W \leftarrow \text{Chol}(C_r^{-1}) W$ where $\text{Chol}(\cdot)$ denotes the upper triangular Cholesky factorization divided by the upper left element.
4. If $\text{Chol}(C_r^{-1})$ is not sufficiently near to identity matrix, then go to step 2.
5. Calculate the shape matrix estimate $D = (d/\text{tr}((W^T W)^{-1}))(W^T W)^{-1}$.
6. Calculate the test statistics using the transformed observations $W \mathbf{y}_{ij}$.
7. Calculate the adjusted treatment effect estimates using the transformed observations $W \mathbf{y}_{ij}$. The final estimates are then $W^{-1} \tilde{A}_{jj'}$, $j' = 1, \dots, k$

Table 4

Bean plants data (Seber, 1984)

Blocks	Treatments					
	1	2	3	4	5	6
1	1.7	1.7	1.4	0.1	1.3	1.7
	0.4	1.0	0.8	0.8	1.0	0.5
	0.20	0.40	0.28	0.10	0.12	0.74
2	1.2	1.2	1.5	0.2	1.4	2.1
	1.4	0.6	0.8	1.2	1.2	1.0
	0.20	0.25	0.83	0.08	0.20	0.59
3	1.3	1.7	1.1	0.3	1.3	2.3
	0.6	0.1	0.7	1.2	0.8	0.4
	0.36	0.32	0.58	0.00	0.30	0.50
4	1.7	1.1	1.1	0.0	1.2	1.3
	1.1	0.0	0.9	0.4	0.6	0.9
	0.39	0.29	0.50	0.00	0.36	0.28

Table 5

Bean plants data: spatial ranks

Blocks	Treatments						
	1	2	3	4	5	6	Sum
1	0.35	0.43	−0.08	−0.81	−0.20	0.31	0.00
	−0.50	0.40	0.00	0.02	0.38	−0.31	0.00
	−0.21	0.14	−0.03	−0.13	−0.32	0.56	0.00
2	−0.15	−0.13	0.14	−0.77	0.17	0.73	0.00
	0.49	−0.52	−0.24	0.11	0.17	−0.01	0.00
	−0.17	−0.15	0.53	−0.18	−0.21	0.18	0.00
3	−0.02	0.28	−0.35	−0.69	0.01	0.77	0.00
	−0.13	−0.56	0.08	0.38	0.35	−0.12	0.00
	−0.03	−0.07	0.39	−0.25	−0.17	0.12	0.00
4	0.65	0.01	−0.18	−0.75	0.06	0.21	0.00
	0.45	−0.69	0.32	−0.16	−0.23	0.30	0.00
	0.07	−0.03	0.32	−0.22	0.03	−0.17	0.00
Sum	0.83	0.60	−0.47	−3.01	0.04	2.02	0.00
	0.31	−1.37	0.17	0.36	0.67	−0.14	0.00
	−0.35	−0.11	1.22	−0.78	−0.67	0.69	0.00

To illustrate the effect of rescaling the marginal responses we multiplied the second component by 1000 (i.e. the weight of beans per plot is in grams). See [Table 6](#) for the test results. We can see that the results are quite similar for the permutation

Table 6
MANOVA test and multivariate Friedman test for the bean plants data

Wt of beans measured in	Test statistic	Value of test. stat.	Asym. <i>p</i> -value	Perm. <i>p</i> -value
kg or g	Q (aff.)	32.10	0.006	0.00075
kg or g	Q_r (aff.)	30.75	0.009	0.0018
kg	Q_r	30.48	0.010	0.0019
g	Q_r	14.10	0.518	0.570

The weight of beans per plot was measured both in kilograms (kg) and in grams (g). Q (aff.) refers to the affine invariant permutation MANOVA test. Q_r and Q_r (aff.) refer to the Friedman test and the affine invariant version of the Friedman test, correspondingly. The asymptotic *p*-values were calculated using χ^2_{15} distribution. The permutation test *p*-values are based on 100,000 randomizations.

MANOVA test and the affine invariant version of Friedman test. Furthermore, the rescaling of the second component seems to have disastrous effect on the noninvariant multivariate Friedman test.

To illustrate the robustness of the Friedman test statistic as compared to the classical MANOVA, we did the following study. We assumed that the response variable is bivariate ($d = 2$), the number of treatments is $k = 3$ and the number of blocks is $n = 8$. We sampled $N = nk = 24$ independent observations y_{ij} from bivariate normal distribution $N(\mu_j, I_2)$ where the expectation was $\mu_j = (0, 0)^T$ for treatments $j = 1, 3$ and $\mu_j = (0, 4)^T$ for treatment $j = 2$. Next, we replaced the observation vector $y_{11} = (y_{111}, y_{112})^T$ by a “contaminated” vector $y_{11} = (0, y_c)^T$ and computed the *p*-values of the MANOVA test and the Friedman test for the various contamination values $y_c = -20, -19, \dots, 19, 20$. By repeating the above procedure for 1000 times we got approximate mean *p*-values for different contamination points y_c . Fig. 1 then illustrates this effect of an outlier y_c on the mean *p*-value. We can see that when the contamination level is low (y_c is near to 0) the MANOVA and Friedman tests give (on average) essentially the same *p*-values and when the contamination level is high the MANOVA test gives much larger *p*-values than the Friedman test.

In the paper, multivariate extensions of the Friedman and Page tests based on the centered spatial ranks were introduced. Also related adjusted and unadjusted treatment effect estimates were considered. The tests and estimates are rotation invariant/equivariant but not scale invariant/equivariant. Affine invariant/equivariant version can however be easily constructed using the transformation and retransformation technique. Another possibility is to use the marginal centered ranks as in [13]. This approach is scale invariant/equivariant but not rotation invariant/equivariant, and the efficiency can again be really poor if the response variables are highly correlated. For constructing affine invariant versions, the transformation and retransformation approach as given in this paper is not then applicable, however, and a different, slightly more complicated technique described in detail by Chakraborty and Chaudhury [1] is needed. Still one possibility is to use affine equivariant centered ranks based on the Oja criterion, see [10]. To compute

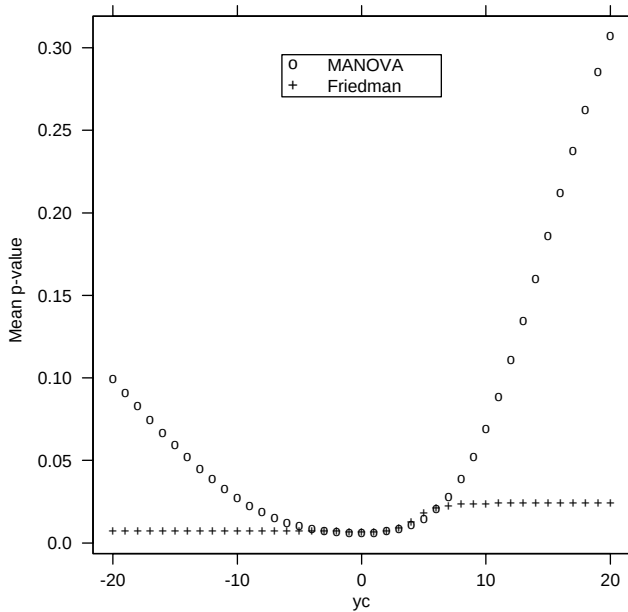


Fig. 1. Mean p -value of the Friedman test and MANOVA test as a function of contamination value.

blockwise centered ranks one has to require, however, that $k \geq d + 1$ which may be a serious limitation.

In the paper it was assumed that the size of each block is equal to the number of the treatments k (complete blocks). In practice, however, it is not always possible to have blocks of that size, and not every treatment can be applied in each block (incomplete blocks). On the other hand, it may also occur that the number of subjects available in blocks may exceed the number of treatments. The multivariate Friedman and Page permutation tests can similarly be constructed also for the situations in which the number n_{ij} of observations on the j th treatment in the i th block is not necessarily equal to 1.

Acknowledgments

The authors wish to thank the Editor and two anonymous referees for careful reading and for valuable comments which greatly improved the presentation.

Appendix. Proofs of the results

Proof of Theorem 2.1. Let $\mathbf{a}$ be an arbitrary $d(k-1) \times 1$ constant vector and

$$Y_n = \mathbf{a}^T \mathbf{z}^* / \sqrt{n} = \sum_{i=1}^n U_i,$$

where $U_i = \mathbf{a}^T \mathbf{z}_i^* / \sqrt{n}$. Now $E^*(U_i) = 0$ and since the second and third moments of $\mathbf{z}_i^*$ are uniformly bounded,

$$\begin{aligned} \text{Var}^*(Y_n) &= \sum_{i=1}^n E^*(U_i^2) = \frac{1}{n} \mathbf{a}^T \text{Var}^*(\mathbf{z}^*) \mathbf{a} \\ &= \mathbf{a}^T \left[\frac{1}{k-1} (kI_{k-1} - J_{k-1}) \otimes C \right] \mathbf{a} \\ &\rightarrow \mathbf{a}^T \left[\frac{1}{k-1} (kI_{k-1} - J_{k-1}) \otimes \Sigma \right] \mathbf{a} \end{aligned}$$

and

$$\begin{aligned} \sum_{i=1}^n E^*(|U_i|^3) &= \sum_{i=1}^n E^*(|\mathbf{a}^T \mathbf{z}_i^* / \sqrt{n}|^3) = \sum_{i=1}^n E^* \left(n^{-3/2} \left| \sum_{j=1}^{k-1} \sum_{l=1}^d a_{jl} z_{ijl} \right|^3 \right) \\ &\leq n^{-1/2} \frac{1}{n} \sum_{i=1}^n E^* \left[\left(\sum_{j=1}^{k-1} \sum_{l=1}^d |a_{jl}| |z_{ijl}| \right)^3 \right] = n^{-1/2} O(1). \end{aligned}$$

Since now

$$\frac{\left(\sum_{i=1}^n E^*(|U_i|^3) \right)^{1/3}}{\left(\sum_{i=1}^n E^*(U_i^2) \right)^{1/2}} \rightarrow 0,$$

the CLT under Liapounoff's condition gives us

$$Y_n \xrightarrow{d} N_1 \left(0, \mathbf{a}^T \left[\frac{1}{k-1} (kI_{k-1} - J_{k-1}) \otimes \Sigma \right] \mathbf{a} \right)$$

which further implies that

$$\mathbf{z}^* / \sqrt{n} \xrightarrow{d} N_{d(k-1)} \left(\mathbf{0}, \frac{1}{k-1} (kI_{k-1} - J_{k-1}) \otimes \Sigma \right). \quad \square$$

Proof of Theorem 2.2. (A) We can use exactly the same procedure as in the proof of Theorem 2.1. Now $Y_n = \mathbf{a}^T \mathbf{r}^* / \sqrt{n} = \sum_{i=1}^n U_i$ and because $|r_{ijl}| \leq 1$ we have $\text{Var}^*(Y_n) \rightarrow \mathbf{a}^T \left[\frac{1}{k-1} (kI_{k-1} - J_{k-1}) \otimes \Sigma_r \right] \mathbf{a}$ and $\sum_{i=1}^n E^*(|U_i|^3) \leq qn^{-1/2} O(1)$. Using the CLT under Liapounoff's condition we get the result

$$\mathbf{r}^* / \sqrt{n} \xrightarrow{d} N_{d(k-1)} \left(\mathbf{0}, \frac{1}{k-1} (kI_{k-1} - J_{k-1}) \otimes \Sigma_r \right). \quad \square$$

(B) The test statistic $\mathbf{L}^*$ can be written as

$$\mathbf{L}^* = -[(k-1, k-2, \dots, 1) \otimes I_d] \mathbf{r}^*.$$

The proof of (A) then implies that the limiting distribution of $n^{-1/2}\mathbf{L}^*$ is a d -variate normal distribution with mean vector $\mathbf{0}$ and covariance matrix

$$\begin{aligned} & [(k-1, \dots, 1) \otimes I_d] \left[\frac{1}{k-1} (kI_{k-1} - J_{k-1}) \otimes \Sigma_r \right] [(k-1, \dots, 1)^T \otimes I_d] \\ &= \frac{1}{k-1} \left(k \sum_{i=1}^{k-1} i^2 - \frac{k(k-1)}{2} \sum_{i=1}^{k-1} i \right) \Sigma_r = \frac{k^2(k+1)}{12} \Sigma_r. \quad \square \end{aligned}$$

Proof of Lemma 2.1. It can be easily seen that the d first elements of $U\mathbf{1}_k$ are $\sum_{j=1}^k T_{1j} = \sum_{i=1}^n \sum_{j=1}^k \text{sgn}(\mathbf{y}_{ij} - \mathbf{y}_{i1}) = -k\mathbf{r}_{.1}$, the next d elements are $-k\mathbf{r}_{.2}$ and so on. $\square$

Proof of Theorem 3.1. Let $j \neq j'$. Since

$$\begin{aligned} \mathbf{0} &= E(\mathbf{z}_{ij}\mathbf{z}_{i1}^T + \dots + \mathbf{z}_{ik}^T) = E(\mathbf{z}_{ij}\mathbf{z}_{i1}^T) + \dots + E(\mathbf{z}_{ij}\mathbf{z}_{ik}^T) \\ &= \Sigma + (k-1)E(\mathbf{z}_{ij}\mathbf{z}_{ij'}^T), \end{aligned}$$

we get $E(\mathbf{z}_{ij}\mathbf{z}_{ij'}^T) = -\frac{1}{k-1}\Sigma$ and thus

$$\text{Var}(\mathbf{z}_i) = E(\mathbf{z}_i\mathbf{z}_i^T) = \frac{1}{k-1} (kI_{k-1} - J_{k-1}) \otimes \Sigma.$$

Under the alternative sequences the test statistic $n^{-1/2} \sum_{i=1}^n \mathbf{z}_{ij}$ is asymptotically d -variate normal with mean vector δ_j and covariance matrix Σ and $n^{-1/2}\mathbf{z}$ is asymptotically $d(k-1)$ -variate normal with mean vector δ and covariance matrix $\frac{1}{k-1}(kI_{k-1} - J_{k-1}) \otimes \Sigma$. $\square$

Proof of Theorem 3.2. We can easily see that

$$\text{Var}(\mathbf{r}_i) = E(\mathbf{r}_i\mathbf{r}_i^T) = \frac{1}{k-1} (kI_{k-1} - J_{k-1}) \otimes B$$

(see the proof of Theorem 3.1).

As earlier, let

$$\mathbf{T}_{jj'} = \sum_{i=1}^n \text{sgn}(\mathbf{y}_{ij'} - \mathbf{y}_{ij})$$

denote the pairwise sign test statistic for comparing the j th and the j' th treatment. Under the alternative sequences the test statistic $n^{-1/2}\mathbf{T}_{jj'}$ is asymptotically d -variate normal with mean vector $A(\delta_{j'} - \delta_j)$ and covariance matrix B [7] and

$$n^{-1/2}\mathbf{r} = \begin{pmatrix} \frac{1}{k} \sum_{s=1}^k n^{-1/2} \mathbf{T}_{s1} \\ \vdots \\ \frac{1}{k} \sum_{s=1}^k n^{-1/2} \mathbf{T}_{s,k-1} \end{pmatrix}$$

is asymptotically $d(k-1)$ -variate normal with mean vector

$$\begin{pmatrix} \frac{1}{k} \sum_{s=1}^k A(\delta_1 - \delta_s) \\ \vdots \\ \frac{1}{k} \sum_{s=1}^k A(\delta_{k-1} - \delta_s) \end{pmatrix} = \begin{pmatrix} A\delta_1 \\ \vdots \\ A\delta_{k-1} \end{pmatrix} = (I_{k-1} \otimes A)\delta$$

and covariance matrix $\frac{1}{k-1}(kI_{k-1} - J_{k-1}) \otimes B$. $\square$

Proof of Corollary 3.2. (A) Using Theorem 3.2

$$Q = (n^{-1/2}\mathbf{r})^T (\text{Var}(\mathbf{r}_i))^{-1} (n^{-1/2}\mathbf{r}) \xrightarrow{d} \chi_{d(k-1)}^2(\xi),$$

where

$$\begin{aligned} \xi &= [(I_{k-1} \otimes A)\delta]^T \left[\frac{1}{k-1} (kI_{k-1} - J_{k-1}) \otimes B \right]^{-1} [(I_{k-1} \otimes A)\delta] \\ &= \delta^T (I_{k-1} \otimes A^T) \left[\frac{k-1}{k} (I_{k-1} + J_{k-1}) \otimes B^{-1} \right] (I_{k-1} \otimes A)\delta \\ &= \frac{k-1}{k} \sum_{i=1}^k \delta_i^T (A^T B^{-1} A) \delta_i. \quad \square \end{aligned}$$

(b) The test statistic $\mathbf{L}$ can be written as

$$\mathbf{L} = -[(k-1, k-2, \dots, 1) \otimes I_d] \mathbf{r}.$$

Theorem 3.2 then implies that the limiting distribution of $n^{-1/2}\mathbf{L}$ is a d -variate normal distribution with mean vector

$$\begin{aligned} &-[(k-1, k-2, \dots, 1) \otimes I_d] (I_{k-1} \otimes A) \delta \\ &= -\sum_{i=1}^{k-1} (k-i) A \delta_i = A \sum_{i=1}^k i \delta_i \end{aligned}$$

and covariance matrix $\frac{k^2(k+1)}{12} B$ (see proof of Theorem 2.2(B)). We then get

$$\begin{aligned} n^{-1/2} \mathbf{L}^T \left(\frac{k^2(k+1)}{12} C_r \right)^{-1} n^{-1/2} \mathbf{L} &= \frac{12}{nk^2(k+1)} \mathbf{L}^T C_r^{-1} \mathbf{L} \\ &\xrightarrow{d} \chi_d^2 \left(\left(\sum_{j=1}^k j \delta_j^T \right) A^T \left(\frac{k^2(k+1)}{12} B \right)^{-1} A \left(\sum_{j=1}^k j \delta_j \right) \right) \\ &= \chi_d^2 \left(\frac{12}{k^2(k+1)} \left(\sum_{j=1}^k j \delta_j^T \right) A^T B^{-1} A \left(\sum_{j=1}^k j \delta_j \right) \right). \quad \square \end{aligned}$$

Proof of Corollary 3.3. It can be seen, using Stirling's formula and the asymptotic expansion of the Gauss hypergeometric function, that

$$\lim_{d \rightarrow \infty} \frac{d\Gamma^2(\frac{d}{2})}{\Gamma^2(\frac{d+1}{2})} = 2 \quad \text{and} \quad \lim_{d \rightarrow \infty} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; \frac{d+2}{2}; \frac{1}{4}\right) = 1. \quad \square$$

Proof of Corollary 3.4. It can be shown, using Stirling's formula for the gamma function, that for large d

$$\text{ARE}_{12} \sim 2a_{k,v} \left[1 + b_{k,v} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} c_{d,v,i,j} \right]^{-1},$$

where

$$a_{k,v} = \frac{\Gamma^2(\frac{2v+1}{2})\Gamma^4(\frac{v+1}{2})k}{(v-2)\Gamma^4(\frac{v}{2})(v!)^2}, \quad b_{k,v} = \frac{\Gamma(\frac{3v}{2})(k-2)}{2^{\frac{v}{2}}\Gamma^3(\frac{v}{2})}$$

and

$$c_{d,v,i,j} = \frac{d^{\frac{v}{2}}\Gamma(\frac{v+1}{2}+i)\Gamma(\frac{v+1}{2}+j)}{(\frac{d}{2}+i)^{\frac{1}{2}}(\frac{d}{2}+j)^{\frac{1}{2}}(\frac{d}{2}+i+j)^{\frac{3v}{2}}i!j!}.$$

Thus it remains to investigate the limit of the double sum. The terms of the double sum, contributing to the limit, are from $i, j \geq \varepsilon d$, for any small $\varepsilon > 0$. We get for the (i, j) th term, using Stirling's formula,

$$c_{d,v,i,j} \sim \frac{d^{\frac{v}{2}}(\frac{v+1}{2}+i)^{\frac{v-1}{2}}(\frac{v+1}{2}+j)^{\frac{v-1}{2}}}{(\frac{d}{2}+i)^{\frac{1}{2}}(\frac{d}{2}+j)^{\frac{1}{2}}(\frac{d}{2}+i+j)^{\frac{3v}{2}}}.$$

The double sum can be approximated by a double integral. Transforming $i = ud$, $j = vd$, $u > \varepsilon$, $v > \varepsilon$, the double sum is asymptotically equal to

$$\begin{aligned} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} c_{d,v,i,j} &\sim \int_{\varepsilon}^{\infty} \int_{\varepsilon}^{\infty} \frac{(uv)^{\frac{v-1}{2}}(1+o(1))}{(\frac{1}{2}+u)^{\frac{1}{2}}(\frac{1}{2}+v)^{\frac{1}{2}}(\frac{1}{2}+u+v)^{\frac{3v}{2}}} du dv \\ &\sim \int_0^{\infty} \int_0^{\infty} \frac{(uv)^{\frac{v-1}{2}}}{(\frac{1}{2}+u)^{\frac{1}{2}}(\frac{1}{2}+v)^{\frac{1}{2}}(\frac{1}{2}+u+v)^{\frac{3v}{2}}} du dv \end{aligned}$$

for $\varepsilon \rightarrow 0$. $\square$

References

- [1] B. Chakraborty, P. Chaudhuri, On affine equivariant sign and rank tests in one and two sample multivariate problems, in: S. Ghosh (Ed.), *Multivariate Analysis, Design of Experiments and Survey Sampling*, Marcel Dekker, New York, 1999.
- [2] T.P. Hettmansperger, J.W. McKean, *Robust Nonparametric Statistical Methods*, Arnold, London, 1998.
- [3] T.P. Hettmansperger, R. Randles, A practical affine equivariant multivariate median, 2003, to appear.
- [4] M. Hollander, D.A. Wolfe, *Nonparametric Statistical Methods*, Wiley, New York, 1973.
- [5] E.L. Lehmann, *Nonparametrics: Statistical Methods Based on Ranks*, Revised, 1st Edition, Prentice-Hall, Englewood Cliffs, NJ, 1998.
- [6] J. Möttönen, H. Oja, Multivariate spatial sign and rank methods, *J. Nonparam. Statist.* 5 (1995) 201–213.
- [7] J. Möttönen, H. Oja, J. Tienari, On the efficiency of multivariate spatial sign and rank tests, *Ann. Statist.* 25 (1997) 542–552.
- [8] A. Niinimaa, H. Oja, Multivariate median, in: S. Kotz, Read, Campbell, D.L. Banks (Eds.), *Encyclopedia of Statistical Sciences*, Vol. 3, Update, Wiley, New York, 1999, pp. 497–505.
- [9] J. Nyblom, T. Mäkeläinen, Comparisons of tests for the presence of random walk coefficients in a simple linear model, *J. Amer. Statist. Assoc.* 78 (1983) 856–864.
- [10] H. Oja, Affine invariant multivariate sign and rank tests and corresponding estimates: a review, *Scand. J. Statist.* 26 (1999) 319–343.
- [11] M.L. Puri, P.K. Sen, *Nonparametric Methods in Multivariate Analysis*, Wiley, New York, 1971.
- [12] P.J. Rousseeuw, S. Van Aelst, K. Van Driessen, Robust multivariate regression, 2003, submitted.
- [13] G.A.F. Seber, *Multivariate Observations*, Wiley, New York, 1984.
- [14] D.E. Tyler, A distribution-free M -estimator of multivariate scatter, *Ann. Statist.* 15 (1987) 234–251.
- [15] S. Visuri, V. Koivunen, H. Oja, Sign and rank covariance matrices, *J. Statist. Plann. Inference* 91 (2000) 557–575.